

HDOC Day-14

Question: Menu-Driven Program for Digit Uniqueness and Frequency Analysis

1. Problem Statement & Identification

- **Problem:** Create a menu-driven program using switch-case to input a positive integer and perform one of the following operations:
 1. Check whether all digits of the integer are unique.
 2. Find the frequency of occurrence for each digit from 0 to 9.
- **Input:** A positive integer (num) and menu choice (choice).
- **Output:** "Unique" or "Not Unique" for Option 1, or the frequency count freq_0 through freq_9 for Option 2.

2. Algorithm

1. **Start**
2. Input a positive integer num.
3. Display menu options (1 and 2).
4. Read user choice.

5. Execute operations based on choice using switch-case:

- **Case 1:**

1. Initialize a frequency array `freq[10]` with zeros and flag `is_unique = 1`.
2. Set `temp = num`.
3. While `temp > 0`:
 - Extract digit `digit = temp % 10`.
 - Increment `freq[digit]`.
 - If `freq[digit] > 1`, set `is_unique = 0` and break out of loop.
 - Update `temp = temp / 10`.
4. If `is_unique == 1`, print "Unique", else print "Not Unique".

- **Case 2:**

1. Initialize frequency array `freq[10]` with zeros.
2. Set `temp = num`.
3. While `temp > 0`:
 - Extract digit `digit = temp % 10`.
 - Increment `freq[digit]`.
 - Update `temp = temp / 10`.
4. Loop `i` from 0 to 9 and print `freq_i = freq[i]`.

- **Default:** Print "Invalid Choice!".

6. **End**

3. Test Case Table

Tes t Ca se	Input Num ber	Choi ce	Operati on	Expec ted Outpu t	Actu al Out put	Stat us
1	1234 5	1	Check Unique ness	Uniqu e	Uniq ue	Pas s
2	1223 4	1	Check Unique ness	Not Uniqu e	Not Uniq ue	Pas s
3	1123 33	2	Digit Freque ncy	freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3, freq_4 = 0,	freq _0 = 0, freq _1 = 2, freq _2 = 1, freq	Pas s

Test Case	Input Number	Choice	Operation	Expected Output	Actual Output	Status
				freq_5 = 0, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0	_3 = 3, freq_4 = 0, freq_5 = 0, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0	

Test Case	Input Number	Choice	Operation	Expected Output	Actual Output	Status
4	9080	1	Check Uniqueness	Not Unique	Not Unique	Pass

4. C Source Code

```
#include <stdio.h>

int main(void) {
    int choice, num, temp, digit;
    int freq[10] = {0};
    int is_unique = 1;

    printf("Enter a positive integer: ");
    if (scanf("%d", &num) != 1 || num <= 0) {
        printf("Invalid input! Please enter a positive integer.\n");
        return 1;
    }

    printf("\n--- MENU ---\n");
    printf("1. Check whether all its digits are unique\n");
    printf("2. Find the frequency of each digit from '0' to '9'\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            temp = num;
            while (temp > 0) {
                digit = temp % 10;
                freq[digit]++;
            }
        }
    }
```

```
        if (freq[digit] > 1) {
            is_unique = 0;
            break;
        }
        temp /= 10;
    }

    if (is_unique) {
        printf("\nNumber: %d, Output: Unique\n", num);
    } else {
        printf("\nNumber: %d, Output: Not Unique\n", num);
    }
    break;
}
```

case 2:

```
temp = num;
while (temp > 0) {
    digit = temp % 10;
    freq[digit]++;
    temp /= 10;
}

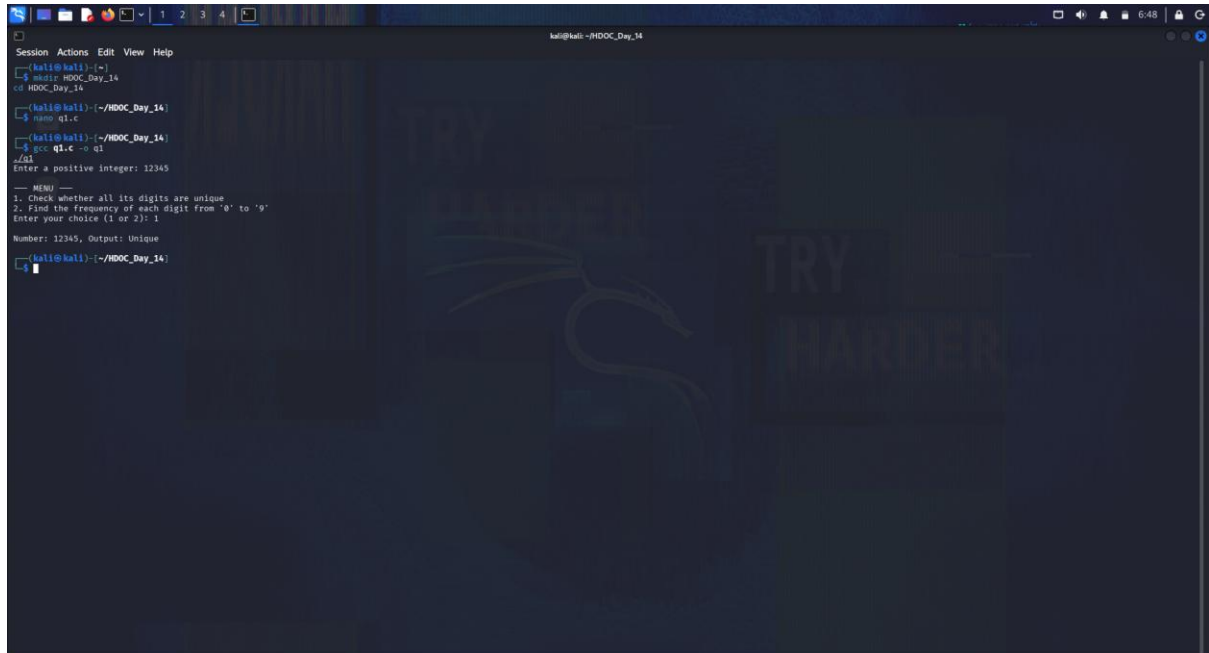
printf("\nNumber: %d\n", num);
for (int i = 0; i <= 9; i++) {
    printf("freq_%d = %d", i, freq[i]);
    if (i < 9) {
        printf(", ");
    }
}
printf("\n");
break;
}
```

default:

```
    printf("\nInvalid Choice!\n");
    return 1;
}

return 0;
}
```

5. Compilation, Execution & Screenshots



```
kali@kali:~/HDOOC_Day_14$ cd HDOOC_Day_14
kali@kali:~/HDOOC_Day_14$ nano q1.c
kali@kali:~/HDOOC_Day_14$ gcc q1.c -o q1
./q1
Enter a positive integer: 12345

--- MENU ---
1. Check whether all its digits are unique
2. Find the frequency of each digit from '0' to '9'
Enter your choice (1 or 2): 1

Number: 12345, Output: Unique
kali@kali:~/HDOOC_Day_14$
```

The screenshot shows a terminal window with a dark background and light-colored text. The user is in the directory ~/HDOOC_Day_14. They have created a file named q1.c using nano and compiled it with gcc to produce an executable named q1. They then run the program, which prompts for a positive integer. The user enters 12345. The program displays a menu with two options: 1. Check whether all its digits are unique, and 2. Find the frequency of each digit from '0' to '9'. The user selects option 1, and the program outputs "Number: 12345, Output: Unique".